MPC-PiNNs technique applied to Quadrotors

Elective in Robotics [UAV] Course Project

Master's Degree in Artificial Intelligence and Robotics

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Neural Model Predictive Control (NMPC)

1 Introduction

In this work we have tried to combine Model Predictive Control along with PiNNs to trajectory tracking problem applied to quadrotors. We have used the Unicycle robot as simple test case for our experiments.

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2 Quadrotor Model

Quadrotor Standard Model

Starting from the Translational (1) and Rotational Dynamics (2), we have derived the simplified second order of the quadrotor.

$$m\ddot{p} = f_t + f_g + f_a + f_d \tag{1}$$

$$J\dot{\omega} + \omega \times (J\omega) = \tau \tag{2}$$

$$\begin{cases} \ddot{x} = -\Big(\cos\psi\ \sin\theta\ \cos\phi + \sin\psi\ \sin\phi\Big)\frac{T}{m}, \\ \ddot{y} = -\Big(\sin\psi\ \sin\theta\ \cos\phi - \cos\psi\ \sin\phi\Big)\frac{T}{m}, \\ \ddot{z} = -\Big(\cos\theta\ \cos\phi\Big)\dfrac{T}{m} + g, \\ \ddot{\phi} = \frac{\tau_\phi}{I_x} \\ \ddot{\theta} = \frac{\tau_\theta}{I_y} \\ \ddot{\psi} = \frac{\tau_\psi}{I_z} \end{cases} \tag{3}$$

Crazyflie Model

2 Quadrotor Model

Since we have used the Crazyflie as reference, the reference model is different with respect to the previous one. This is a consequence of how the firmware of the robot acts. There is a low level attitude controller that takes as inputs the RPY desired poses and maps them as control torques.

$$
\begin{cases}
\dot{x} = v_x, \\
\dot{y} = v_y, \\
\dot{z} = v_z, \\
\dot{v}_x = -\Big(\cos\psi\ \sin\theta\ \cos\phi + \sin\psi\ \sin\phi\Big)\frac{T}{m}, \\
\dot{v}_y = -\Big(\sin\psi\ \sin\theta\ \cos\phi - \cos\psi\ \sin\phi\Big)\frac{T}{m}, \\
\dot{v}_z = -\Big(\cos\theta\ \cos\phi\Big)\dfrac{T}{m} + g, \\
\dot{\phi} = (\phi_c - \phi)/\tau \\
\dot{\theta} = (\theta_c - \theta)/\tau \\
\dot{\psi} = (\psi_c - \psi)/\tau
\end{cases}
\tag{4}
$$

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MPC Structure

3 Model Predictive Control

We have used for both robots the following structure for the MPC:

$$\min_{u(0),u(1),\ldots,u(N-1)} \quad \sum_{k=0}^{N-1} \ell(x(k),u(k)) + \ell_f(x(N)) \tag{5}$$

$$\text{s.t.} \quad x(k+1) = f(x(k),u(k)), \quad k = 0,\ldots,N-1, \tag{6}$$

$$x(k) \in \mathcal{X}, \quad u(k) \in \mathcal{U}, \quad k = 0,\ldots,N-1, \tag{7}$$

$$x(0) = x_{0.} \tag{8}$$

MPC Cost Function

Cost Function

$$\ell(x(k), u(k)) = \sum_{k=0}^{N-1} \left[\sum_{x \in \mathcal{X}} W_{x(k)} \left(x(k)_{\text{ref}} - x(k) \right)^2 + \sum_{u \in \mathcal{U}} W_u \, u^2 \right],$$

$$\ell_f(x(N)) = \sum_{x \in \mathcal{X}} W_{x(N)} \left(x(N)_{\text{ref}} - x(N) \right)^2.$$

Unicycle State & Control

$$\begin{cases} \mathcal{X} = \begin{bmatrix} x & y & \theta \end{bmatrix}^T, \\ \mathcal{U} = \begin{bmatrix} v & \omega \end{bmatrix}^T. \end{cases}$$

Quadrotor State & Control

$$\begin{cases} \mathcal{X} = \begin{bmatrix} x & y & z & v_x & v_y & v_z & \phi & \theta & \psi \end{bmatrix}^T, \\ \mathcal{U} = \begin{bmatrix} \phi_c & \theta_c & \psi_c & T \end{bmatrix}^T. \end{cases}$$

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4 Dataset

Structure

4 Dataset

The data the we used to train our neural models come from simulation and some real world experiments.

- Generally speaking, we can describe each our dataset as a tuple in the form $\{[x_t, u_t], [x_{t+1}]\}$, were x_t is the state of the system at time t, u_t is the control input at time t and x_{t+1} is the state of the system at time $t + 1$.
- For the baseline (*simple-neural-network*), no constraints on the dataset form are required, while on the physics informed models the input needs to be **constant** in a sub-interval $[0, T]$.

Structure (2)

4 Dataset

The unicycle dataset it's composed by 500 trajectories, each composed by 1000 steps, where the inputs are maintained constant for the whole episode.

Unicycle Input Combinations:

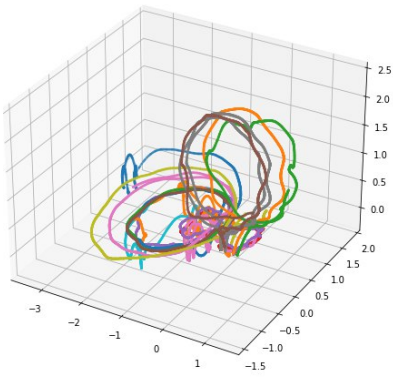
- $v > 0, w > 0$
- $v > 0, w < 0$
- $v < 0, w > 0$
- $v < 0, w < 0$
- $v > 0, w = 0$
- $v < 0, w = 0$
- $v = 0, w > 0$
- $v = 0, w < 0$

For the quadrotor we have extracted data from ROS messages given directly by the robot and the simulation. We have defined 19 different types of trajectories that belong to 3 main categories:

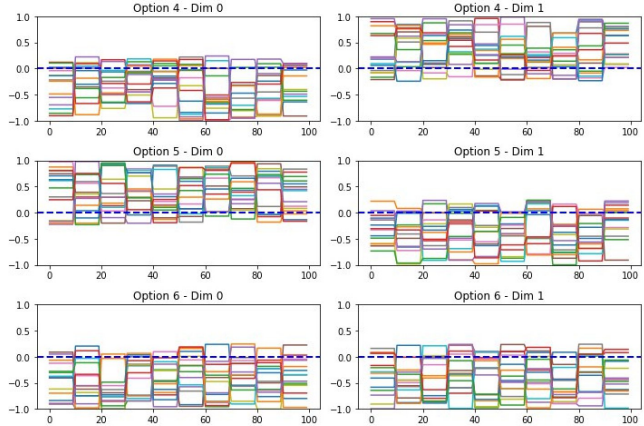
- **horizontal circle**
- **vertical circle**
- **lemniscate**

Examples

4 Dataset



Quadrotor data



Unicycle data (PiNN)

Problems

4 Dataset

When extracting data from the real-world experiments (or the gazebo simulations), state and action **publishing rates** were a crucial aspect to take into account.

- The action publishing rate should be high enough to make the controller safe enough
- The state publishing rate should be high, but it couldn't be too high, for communication latency reasons, which would have made the use of the real crazyflie impossible

These problems could be **partially mitigated** by using the gazebo simulator and setting ad-hoc publishing rates, or forcing the input to be constant (introducing a noisy approximation) for a certain amount of time in the real-world data.

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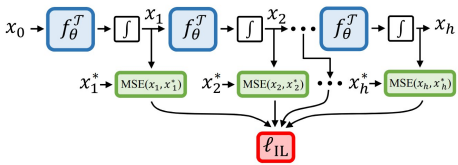
5 Standard Neural Network

Training Scheme

5 Standard Neural Network

The multi-step training procedure is defined as follows:

- Give in input the pair (s_t, a_t) to the network
- Perform the integration of the output in the sample time dt
- Add what we have to the input to obtain the new state s_{t+1}
- Compute the **MSE** loss
- Use the new state as the next input of the network



Unicycle Network

5 Standard Neural Network

The neural network used to model the kinematic model of the unicycle is a simple MLP architecture with two hidden layers of 128 neurons. The input is given by the concatenation of the current state and the current control inputs:

$$X = [x, y, \theta, v, \omega] \tag{9}$$

while the output represents the dynamics of the system:

$$Y = [\dot{x}, \dot{y}, \dot{\theta}] \tag{10}$$

Quadrotor Network

5 Standard Neural Network

The quadrotor dynamics presents a more complex model, with a state $x \in R^9$ and control inputs $u \in R^4$. For approximating such dynamics with the defined training scheme, a larger network has been tested: we have increased the number of hidden layers up to 3 and the latent dimension from 128 to 256. However, the main problem during training is related to the generation of the data.

$$X = \begin{bmatrix} x & y & z & v_x & v_y & v_z & \phi & \theta & \psi \mid U \end{bmatrix}^T \tag{11}$$

$$U = \begin{bmatrix} \phi_c & \theta_c & \psi_c & T \end{bmatrix}^T \tag{12}$$

$$Y = \begin{bmatrix} \dot{x} & \dot{y} & \dot{z} & \dot{v_x} & \dot{v_y} & \dot{v_z} & \dot{\phi} & \dot{\theta} & \dot{\psi} \end{bmatrix}^T \tag{13}$$

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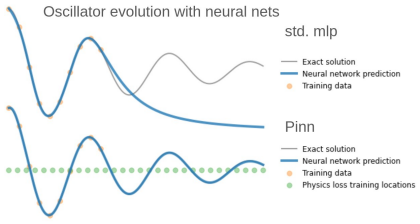
6 Physical Informed Neural Networks

PiNNs

6 Physical Informed Neural Networks

Physical Informed Neural Networks (PiNNs) propose an alternative approach where the solution is approximated by a neural network that is trained not only on data but also by enforcing the underlying physical laws via the differential equations.

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{data}}(\theta) + \mathcal{L}_{\text{phy}}(\theta) \qquad (14)$$



PiNN vs Classical MLP

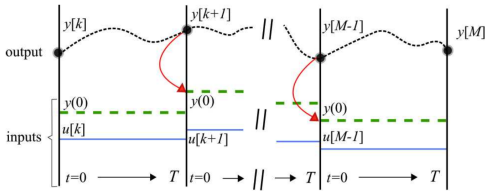
Problems and Limitations:

- Cannot handle control input
- Fixed initial condition

PiNN with Controls

6 Physical Informed Neural Networks

The Pinn-C modifies the formulation of the pinn, by dividing the evolution into sub-transitions with constant input. The network takes as input $(x(0), \bar{u}, t)$.



Pros:

- it can handle larger time-steps with respect to the original formulation
- can accept arbitrary starting conditions

Cons:

- relies heavily on the dataset

Training Scheme

Training Loop: Loss Calculation

1: **Input:** X_{data}, y
2: **Initialize:** $L \leftarrow 0$
3: **Compute Predictions:** $\hat{y} \leftarrow \mathcal{M}(X_{\text{data}})$
4: **Update Loss with MSE:** $L \leftarrow L + \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
5: **Sample Input Data:** $X \sim \mathcal{D}$
6: **Calculate Gradient:** $J \leftarrow \nabla_X \theta$
7: **Compute Analytical ODE:** $\text{ode}(X) = f(X)$
8: **Update Loss with ODE Term:** $L \leftarrow L + \frac{1}{n}\sum_{i=1}^{n}\left(\text{ode}(X) - \nabla_X \theta\right)^2$
9: **Sample a Point:** $X_{\text{point}} \sim \mathcal{D}$
10: **Set to zero the Input:** $X_{\text{input}} \leftarrow 0$
11: **Update Loss with Boundary Term:** $L \leftarrow L + \frac{1}{n}\sum_{i=1}^{n}(X_i - \hat{y}_i)^2$
12: **Output:** Updated loss L

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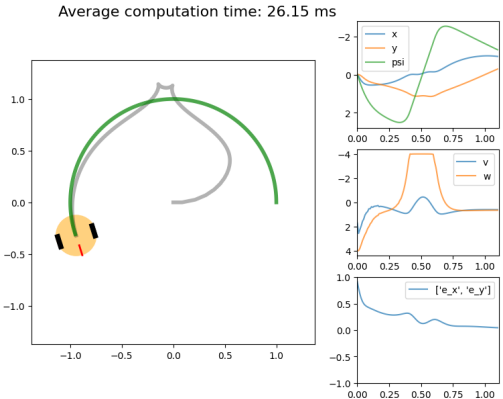
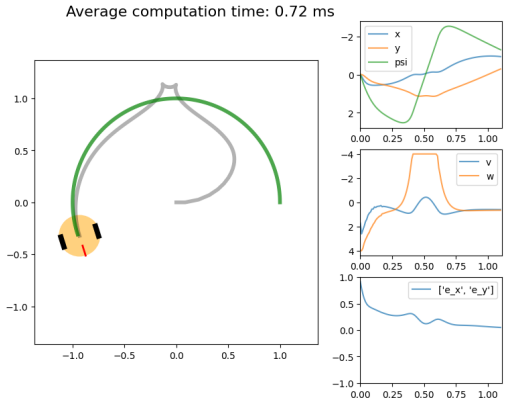
Unicycle Standard Net (1)

7 Results and Future Works



(a) MPC tracking with learned model

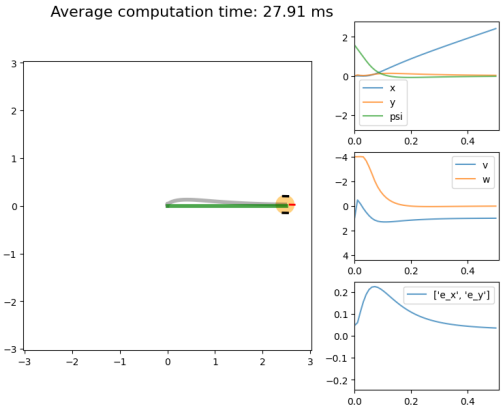


(b) MPC tracking with nominal model

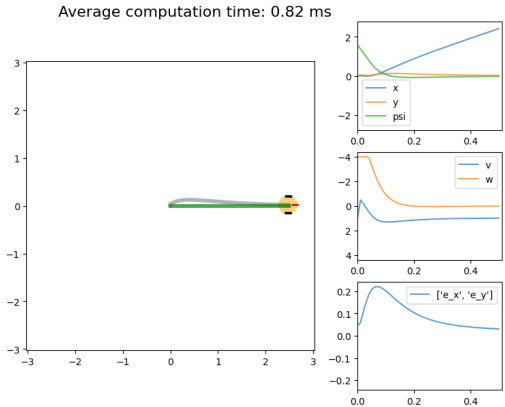
Figure: Circle comparison

Unicycle Standard Net (2)

7 Results and Future Works



(a) Straight line velocity testing with neural model



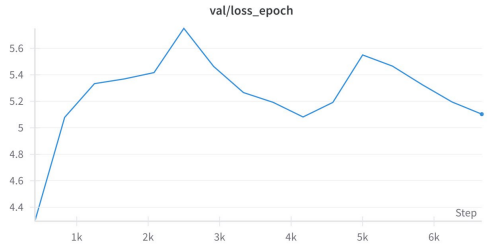
(b) Straight line velocity testing with nominal model

Figure: Straight Line case comparison

Quadrotor Standard Net

7 Results and Future Works

(a) Train Loss

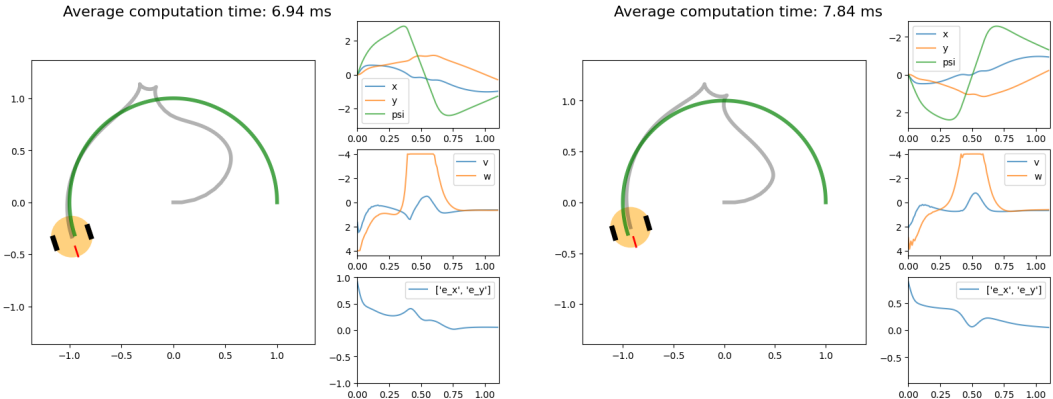


(b) Validation Loss

Figure: Loss Behavior of the Quadrotor case

Unicycle PiNN-C (1)

7 Results and Future Works



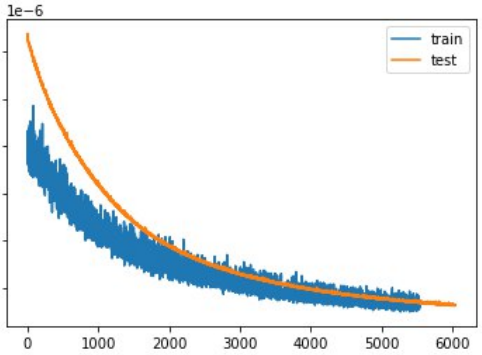
(a) MPC tracking with physics loss

(b) MPC tracking without physics loss

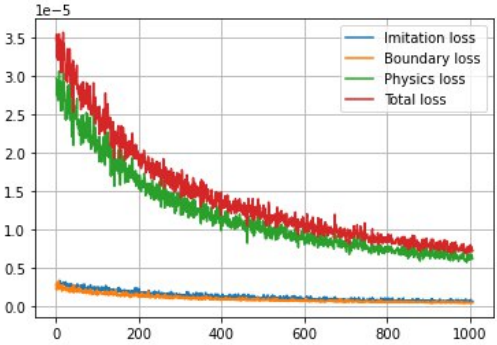
Figure: Pinn prediction evaluation

Unicycle PiNN-C (2)

7 Results and Future Works

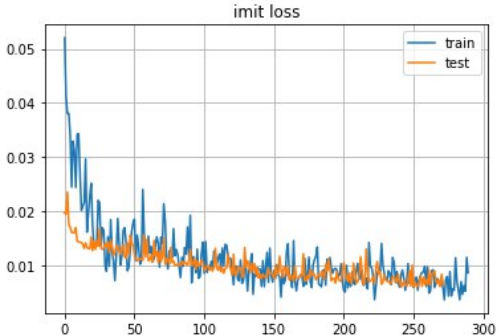


(a) test and train loss for the imit component

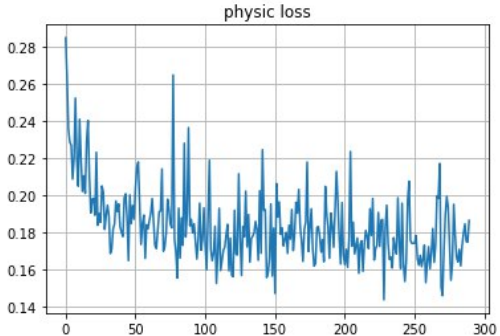


(b) train loss for the pinn-c model

Figure: Pinn loss evaluation for the unicycle



(a) test and train loss for the imit component of the quadrotor model



(b) physics loss of the pinn-c model for the quadrotor

Figure: Pinn loss evaluation for the quadrotor

Future Works

7 Results and Future Works

Despite all the challenges that we have encountered, the results underscore the potential of using physics-informed models along with Model Predictive Control. We have some ideas to improve the performances and solve some of the problems:

- **Reservoir computing** can be used to deal with unbalance and non-homogeneous input signal, while the output layers can be tuned in a physics informed fashion
- **Double stage training** mechanism could be used for linearly connect multiple models
- **Moving average** technique in the training phase could be beneficial for the convergence of the network.

MPC-PiNNs technique applied to Quadrotors

Thank you for listening